

NETWORK LAB REPORT

Layer 2 Security: Port Security Configuration & Troubleshooting

Prepared by: Mohamed Ashraf | Date: August 2, 2026

1. Lab Topology & Addressing

Network Segment: 10.0.0.0/24

Devices:

- **End Devices:** PC1 (.1), PC2 (.2), PC3 (.3)
- **Switches:** SW1 (Access Switch), SW2 (Distribution/Core)
- **Router:** R1 (Default Gateway - 10.0.0.254)

Physical Connections:

- PC1, PC2, and PC3 connect to SW1 via interfaces F0/1, F0/2, and F0/3 respectively.
- SW1 connects to SW2 via a Trunk/Uplink (G0/1 on both ends).
- SW2 connects to R1 via G0/2 on the switch and G0/0 on the router.

2. Switch Configuration

2.1. SW1 Configuration (Access Ports)

Port security is enforced on the access ports connected to the PCs. The interfaces are restricted to a single MAC address, and aging is implemented.

```
SW1(config)# int range f0/1 - 3
SW1(config-if-range)# no shut
SW1(config-if-range)# switchport mode access
SW1(config-if-range)# switchport port-security
SW1(config-if-range)# switchport port-security maximum 1
SW1(config-if-range)# switchport port-security aging time 60
```

2.2. SW2 Configuration (Uplink/Trunk)

Port security is configured on the G0/1 interface. The violation mode is set to `restrict` to drop unauthorized frames without shutting down the vital uplink port.

```
SW2(config)# int g0/1
SW2(config-if)# no shut
SW2(config-if)# switchport mode access
SW2(config-if)# switchport port-security
SW2(config-if)# switchport port-security violation restrict
SW2(config-if)# switchport port-security maximum 4
SW2(config-if)# switchport port-security mac-address sticky
```

3. Validation & Testing

3.1. Initial Connectivity Test

A standard ICMP echo request from PC1 to the default gateway (R1) successfully passes through the network, validating base connectivity.

```
C:\> ping 10.0.0.254

Pinging 10.0.0.254 with 32 bytes of data:
Reply from 10.0.0.254: bytes=32 time=156ms TTL=255
Reply from 10.0.0.254: bytes=32 time=12ms TTL=255
Reply from 10.0.0.254: bytes=32 time<1ms TTL=255
Reply from 10.0.0.254: bytes=32 time<1ms TTL=255
```

3.2. SW2 Port Security Verification

Verifying the port security status on SW2's G0/1 after initial traffic generation. Sticky MAC addresses have been successfully populated into the table up to the configured maximum of 4.

```
SW2# show port-security int g0/1
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Maximum MAC Addresses   : 4
Total MAC Addresses     : 4
Configured MAC Addresses : 0
Sticky MAC Addresses    : 4
Last Source Address:Vlan : 0060.471C.1D19:1
Security Violation Count : 0
```

4. Troubleshooting & Security Violations

4.1. Inducing a Violation (SW1 SVI Configuration)

To trigger a topology change, an SVI (VLAN 1) was configured on SW1 with an IP address (10.0.0.10). When a ping was initiated from SW1 to the Gateway (10.0.0.254), it failed, leading to security violations upstream.

```
SW1(config)# int vlan 1
SW1(config-if)# ip address 10.0.0.10 255.255.255.0
SW1(config-if)# no shut
SW1(config-if)# do ping 10.0.0.254
Success rate is 0 percent (0/5)
```

4.2. SW2 Syslog & Restrict Mode Behavior

As SW2's G0/1 already learned its maximum of 4 MAC addresses, the new traffic generated a violation. Because the mode was set to `restrict`, the port remained operational, but dropped the illicit frames and generated syslog messages.

```
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC
address 0060.7024.2366 on port GigabitEthernet0/1.
```

Checking the port security on SW2 confirms the violation counter incremented:

```
SW2# show port-security int g0/1
Security Violation Count : 6
```

4.3. SW1 Shutdown Mode Behavior

Subsequent pings from PC1 (MAC: 0001.0001.000A) began failing (Request timed out). Inspecting SW1 reveals that interface F0/1 experienced a violation. Since SW1 is running the default shutdown mode, the port entered the **err-disabled** state.

```
SW1# show port-security int f0/1
Port Security : Enabled
Port Status : Secure-shutdown
Violation Mode : Shutdown
Maximum MAC Addresses : 1
Total MAC Addresses : 0
Last Source Address:Vlan : 0001.0001.000A:1
Security Violation Count : 1
```

5. Conclusion

This lab successfully demonstrated the critical differences between port security violation modes in Cisco IOS:

- **Shutdown (SW1):** Immediately disables the port (Secure-shutdown / err-disable) upon detecting an unauthorized MAC, requiring administrative intervention or errdisable recovery.
- **Restrict (SW2):** Keeps the port forwarding for authorized MACs, drops unauthorized traffic, and logs the event (Syslog + counter increment), which is highly preferable for uplink interfaces to prevent widespread outages.